Setting A — HCI: Mercer ED Triage Desk

ENVIRONMENT

AI TRIAGE SUPPORT SYSTEM CO-DESIGN CANVAS

What is the AI's context of operation?

Where

What place?
Does it change?

- Mercer Emergency Department triage desk.
- Shared clinical workstation.
- Busy, high-pressure environment.
- May later be expanded to other hospital departments.

User(s)

Who is using the robot?

- ED triage nurses (primary users).

When

What time of day?
Does it change?

- During every patient triage assessment.
- 24 hours per day.
- Peak demand periods and overnight shifts.

Secondary user(s)

Are there secondary users?
E.g. teachers that help students use a robot.

- Emergency physicians.
- Charge nurses.
- Bed managers.
- Clinical IT staff.
- Hospital governance staff.

Data collection

Does the robot collect data from its environment?

- Patient demographics.
- Presenting complaint.
- Vital signs.
- Manual nurse input.
- Future EHR integration.
- Stores prediction history and override logs.

TRADE-OFF:
More data collection requires more attention to data security.

Simultaneous users

How many users should be able to use the robot simultaneously?

TRADE-OFF:
More simultaenous users requires a more sophisticated robot.

one — few — several

Reason:
Multiple triage nurses may use the system simultaneously, but not dozens at one workstation.

External sensors and actuators

Does the robot use external sensors?
Does it have external actuators, such as lights or limbs?

- Manual vital-sign entry.
- Future integration with BP monitors, pulse oximeters and thermometers.
- No physical actuators.
- Visual alerts only.

Connection to systems

Is the robot connected to external systems, such as software, databases, or other robots?
How does it use these systems?

- Mercer Electronic Health Record.
- Hospital user authentication.
- Clinical audit database.
- Future reporting dashboard.
- Secure hospital network.